

Discrete Mathematics: Concept Checklist

Active-Recall & Self-Testing Revision Guide

1 Logic, Quantifiers, and Proof Methods

▮ Conventions & Exam Pitfalls.

- **Linguistic Conjunction:** In mathematical logic, the word “*but*” functions as conjunction ($\wedge$). Example: “ p is true but q is false” $\equiv p \wedge \neg q$.
- **Inference Declarations:** When doing formal inference, always explicitly state what propositions each variable p, q, r represents.
- **Universal Proofs:** When proving $\forall x \in S$, begin with “*Given an arbitrary $x \in S$* ” or “*Let $x \in S$ be arbitrary*”.

Definitions & Concepts to State 1.1

- **Exclusive OR ($\oplus$):** Write down the full truth table of $p \oplus q$. Express $p \oplus q$ in terms of $\wedge, \vee, \neg$.
- **Conditional Statement ($p \implies q$):** List at least 8 equivalent English phrases of $p \implies q$, including:
 - “ q follows from p ”
 - “ q unless $\neg p$ ”
 - “ p only if q ”, “ p is sufficient for q ”, “ q is necessary for p ”, etc.
- **Variants of Conditional:** State the Converse ($q \implies p$), Contrapositive ($\neg q \implies \neg p$), and Inverse ($\neg p \implies \neg q$). Which are logically equivalent?
- **Quantifiers with Restricted Domains:** State the logical expansion for:

$$\forall x < 0 P(x) \quad \text{and} \quad \exists x > 0 Q(x).$$

- **Nested Quantifiers:** Explain why the order of quantifiers matters. Compare $\forall x \exists y P(x, y)$ vs. $\exists y \forall x P(x, y)$ with an example. (Review Assignment 1 nested quantifier problems).
- **Special Proof Methods:**
 - What is a *vacuous proof*? When does it apply?
 - What is a *trivial proof*?
 - What is an *exhaustive proof* (proof by cases)?

▮ Theorems & Identities to State 1.2

- **Core Logic Laws:** State De Morgan's laws, Distributive laws, Absorption laws, and Conditional equivalence.
- **8 Rules of Inference for Propositional Logic:** List all 8 rules and write their tautological forms:
 - Modus Ponens, Modus Tollens, Hypothetical Syllogism, Disjunctive Syllogism
 - Addition, Simplification, Conjunction, Resolution
- **4 Rules of Inference for Quantified Statements:** State Universal Instantiation (UI), Universal Generalization (UG), Existential Instantiation (EI), and Existential Generalization (EG).

▮ Proof Challenges & Calculations 1.3

- **Quantifier Distribution (Proofs):**
 1. Prove: $\forall x (P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$.
 2. Prove: $\exists x (P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$.
- **Quantifier Distribution (Disproofs & Counterexamples):**
 1. Disprove: $\forall x (P(x) \vee Q(x)) \equiv \forall x P(x) \vee \forall x Q(x)$.
 2. Disprove: $\exists x (P(x) \wedge Q(x)) \equiv \exists x P(x) \wedge \exists x Q(x)$.
- **Set Identity Proof via Logic:** Let $A, B \subseteq U$. Prove that $A \subseteq B \Leftrightarrow \overline{A} \cup B = U$ by translating to propositional equivalences.

2 Sets, Functions, and Cardinality

▮ Conventions & Exam Pitfalls.

- **Natural Numbers Convention:** In this course, $\mathbb{N} = \{0, 1, 2, 3, \dots\}$. $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$.
- **Set $\leftrightarrow$ Logic Translation:** $U \leftrightarrow \mathbf{T}$ (True), $\emptyset \leftrightarrow \mathbf{F}$ (False), $\cup \leftrightarrow \vee$, $\cap \leftrightarrow \wedge$, $\overline{A} \leftrightarrow \neg A$.

Definitions & Concepts to State 2.1

- **Set Terminology:** Define singleton set, proper subset ($A \subsetneq B$), power set $\mathcal{P}(A)$, and Cartesian product $A \times B$.
- **Set Operations:** State formal set-builder definitions of $A \cup B$, $A \cap B$, $A \setminus B$, and $\overline{A}$.
- **Function Classifications:** Define real-valued function vs. integer-valued function.
- **Image & Preimage:** For $f : A \rightarrow B$, define $f(S)$ for $S \subseteq A$, and $f^{-1}(T)$ for $T \subseteq B$.
- **Injectivity, Surjectivity, Bijectivity:** Write the formal logical definitions of:
 - Injective (one-to-one)
 - Surjective (onto)
 - Bijective (one-to-one correspondence)
 - Invertible function and inverse function f^{-1}
- **Floor & Ceiling Characterizations:** State the fundamental bounding inequalities:

$$\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1 \quad \text{and} \quad \lceil x \rceil - 1 < x \leq \lceil x \rceil.$$

- **Rounding Function:** Express $\text{round}(x)$ in terms of $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$.
- **Countability Characterization:** State the three equivalent characterizations for a set $S \neq \emptyset$ to be countable.

▮ Theorems & Identities to State 2.2

- **Function Composition Properties** ($f : A \rightarrow B, g : B \rightarrow C$): State the 4 major composition theorems regarding injectivity and surjectivity of $g \circ f$.
- **Invertibility Theorem:** State the condition for f to be invertible (f invertible $\Leftrightarrow f$ bijective).
- **Injection-Surjection Duality:** Show that for non-empty sets, $(\exists \text{ injection } f : A \rightarrow B) \Leftrightarrow (\exists \text{ surjection } g : B \rightarrow A)$.
- **Countable Operations:** State the closure properties of countable sets under subsets, finite unions, countable unions, and Cartesian products.
- **Cantor's Theorem:** State Cantor's theorem comparing $|A|$ and $|\mathcal{P}(A)|$.

[[Proof Challenges & Calculations 2.3

□ **Composition Inverses Counterexamples:**

1. Give an example where $g \circ f$ is injective, but g is **not** injective.
2. Give an example where $g \circ f$ is surjective, but f is **not** surjective.

□ **Countability Proofs:**

- Prove $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countably infinite.
- Prove $\mathbb{Q}$ is countably infinite.
- Prove $\mathbb{R}$ is uncountable via Cantor's diagonal argument.
- Prove $|A| < |\mathcal{P}(A)|$ for all sets A .

□ **Function Space Bijection:** Let $S \neq \emptyset$ and $F_S = \{f : S \rightarrow \{1, 2\}\}$. Prove there is a bijection between F_S and $\mathcal{P}(S)$.

□ **Cardinality Computations ($|A| = m$):** Evaluate:

- $|\{x \in \mathcal{P}(A) : |x| \leq 1\}|$
- $|\{x \subseteq \mathcal{P}(A) : |x| \leq 1\}|$

□ **Equinumerosity Bijections:** Construct explicit bijections showing:

- $|\mathbb{R}| = |(0, \infty)|$
- $|\mathbb{R}| = |(\sqrt{2}, \infty)|$
- $|\mathbb{R}| = |(0, 1)|$
- $|[0, 1]| = |(0, 1)|$
- $|\mathcal{P}(\mathbb{Z})| = |\mathcal{P}(\mathbb{Z}^+)|$

□ **Subsets & Partitions of Reals / Naturals:**

- Prove or disprove: There exists a countably infinite subset of the set of irrational numbers.
- Describe a partition of $\mathbb{N}$ that divides $\mathbb{N}$ into $\aleph_0$ countably infinite subsets.

▮ **Textbook Exercise Checklist.**

Section	Textbook Problems	Status
Section 2.1	Problem 27	□ Done
Section 2.2	Problem 17	□ Done
Section 2.3	Problems 35, 45	□ Done
Section 2.5	Problem 11	□ Done

3 Algorithms and Asymptotic Analysis

▮ **Conventions & Exam Pitfalls.**

- **Logarithm Notation:** In this course, $\log x = \ln x$ unless specified otherwise.
- **Disproving Big- O Strategy:** To prove $f(x) \neq O(g(x))$, use proof by contradiction: assume witnesses C, k exist, and derive that an unbounded function is bounded.

▮ **Definitions & Concepts to State 3.1**

- **Binary Search:** Explain the binary search mechanism and state its recurrence and asymptotic complexity.
- **Asymptotic Definitions:** State the formal definition of:
 - Big- O notation: $f(x) = O(g(x))$ with witnesses C, k .
 - Big- Ω notation: $f(x) = \Omega(g(x))$.
 - Big- Θ notation: $f(x) = \Theta(g(x))$.
 - Little- o notation: $f(x) = o(g(x))$.

▮ Theorems & Identities to State 3.2

- **Limit Criteria:** State the limit test for O, Ω, Θ, o when $L = \lim_{x \rightarrow \infty} \left| \frac{f(x)}{g(x)} \right|$.
- **Sum & Product Rules:** State the sum rule $(f_1 + f_2)(x) = O(\max(|g_1|, |g_2|))$ and product rule $(f_1 f_2)(x) = O(g_1 g_2)$.
- **Symmetry of Θ :** State why $f(x) = \Theta(g(x)) \Leftrightarrow g(x) = \Theta(f(x))$.
- **Duality of O and Ω :** State why $f(x) = O(g(x)) \Leftrightarrow g(x) = \Omega(f(x))$.

▮ Proof Challenges & Calculations 3.3

- **Witness Finding:** Prove $f(x) = x^2 + 2x + 1$ is $O(x^2)$ by finding explicit constants C and k .
- **Disproving Big- O :** Prove by contradiction that n^2 is not $O(n)$.
- **Growth Comparison Tasks:**
 - Prove that for any $n \in \mathbb{N}$ and $a > 0$, the order of x^n is smaller than e^{ax} ($x^n = o(e^{ax})$).
 - Compare the asymptotic orders of x^4 and $x^3 \log^{100} x$.
 - Prove that $\log(n!) = \Theta(n \log n)$.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 3.2	Problems 3, 4, 9, 20, 47, 67, 71, 73	▫ Done

4 Number Theory and Cryptography

▮ Conventions & Exam Pitfalls.

- **Modular Inverses Requirement:** a has an inverse modulo $m \Leftrightarrow \gcd(a, m) = 1$.
- **Euler's Totient Requirement:** $a^{\varphi(n)} \equiv 1 \pmod{n}$ holds if and only if $\gcd(a, n) = 1$.

Definitions & Concepts to State 4.1

- **Divisibility:** Define $a \mid b$. Define quotient and remainder in the Division Algorithm.
- **Congruence Modulo m :** Define $a \equiv b \pmod{m}$ and congruence class $[a]_m$.
- **Prime Concepts:** Define prime, composite, Mersenne prime, relatively prime, and pairwise relatively prime.
- **Sieve of Eratosthenes:** Briefly explain how the Sieve of Eratosthenes identifies all primes up to N .
- **GCD and LCM:** Write the definitions and characterizations of $\gcd(a, b)$ and $\text{lcm}(a, b)$.
- **Bézout's Theorem:** State Bézout's identity for $\gcd(a, b)$.
- **Euler's Totient Function:** Define $\varphi(n)$. State the formula for $\varphi(n)$ and $\varphi(pq)$ for distinct primes.

▮ Theorems & Identities to State 4.2

- ▣ **Divisibility Properties:** State linear combination divisibility: $a \mid b \wedge a \mid c \implies a \mid (mb + nc)$.
- ▣ **Division Algorithm:** State the formal existence and uniqueness theorem for $a = dq + r$.
- ▣ **Modular Arithmetic Properties:** State addition, multiplication, and cancellation rules modulo m .
- ▣ **Prime Factor Bound:** State the theorem: if n is composite, it has a prime factor $p \leq \sqrt{n}$.
- ▣ **Euclidean Reduction:** State $\gcd(a, b) = \gcd(b, r)$ where $a = bq + r$.
- ▣ **Euclid's Lemma:** If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
- ▣ **Fundamental Theorem of Arithmetic:** State the unique prime factorization theorem.
- ▣ **GCD-LCM Product Theorem:** State $\gcd(a, b) \cdot \text{lcm}(a, b) = |ab|$.
- ▣ **Chinese Remainder Theorem (CRT):** State the general CRT for pairwise coprime moduli $m_1, \dots, m_n$.
- ▣ **Fermat's Little Theorem & Euler's Theorem:** State both theorems.
- ▣ **RSA Cryptosystem Principle:** State the RSA identity $m^{de} \equiv m \pmod{n}$ for $n = pq$ and $de \equiv 1 \pmod{\varphi(n)}$.

▮ Proof Challenges & Calculations 4.3

▫ Number Theory Proofs:

- Prove: If $a, b \neq 0$ and $a \mid b$, then $|a| \leq |b|$.
- Prove: $a \equiv b \pmod{m} \Leftrightarrow (a \bmod m) = (b \bmod m)$.
- Prove that every integer > 1 has a prime factor.
- Prove: If $2^m - 1$ is prime, then m is prime.
- Prove Euclid's Theorem: There are infinitely many primes.
- Prove that if p is prime, then either $p \mid a$ or $\gcd(a, p) = 1$.
- Prove: If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.
- Prove: $\gcd(k, mn) = 1 \Leftrightarrow \gcd(k, m) = 1 \wedge \gcd(k, n) = 1$.
- Prove the Chinese Remainder Theorem.
- Prove $\varphi(pq) = (p-1)(q-1)$ for distinct primes p, q .
- Prove: $\lceil n/k \rceil = \lfloor (n-1)/k \rfloor + 1$ for $n, k \in \mathbb{Z}^+$.

▫ Calculations & Algorithms:

- Use the Euclidean Algorithm to compute $\gcd(1236, 2136)$.
- Find the modular inverse of a number using the Extended Euclidean Algorithm.
- Solve the system via CRT: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$.
- Find all solutions to the quadratic congruence $x^2 \equiv 29 \pmod{35}$.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 4.1	Problems 19, 21, 22, 23	☐ Done
Section 4.3	Problems 32, 43	☐ Done
Section 4.4	Problems 11, 21, 40, 65	☐ Done

5 Induction and Recurrence Relations

▮ Definitions & Concepts to State 5.1

- **Induction Templates:** Outline the standard steps of:
 - Principle of Mathematical Induction (Weak Induction).
 - Principle of Strong Mathematical Induction.
- **Well-Ordering Principle:** State the Well-Ordering Principle for non-negative integers.
- **Linear Recurrences Definitions:** Define Linear Homogeneous Recurrence Relations with Constant Coefficients (LHRRWCC) and their characteristic equations.

▮ Theorems & Identities to State 5.2

- **Distinct Roots Theorem:** State the general solution form for $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ when characteristic roots $r_1, \dots, r_k$ are distinct.
- **Repeated Roots Theorem:** State the solution form when root r_i has multiplicity m_i .
- **Linearity of Recurrence Solutions:** State the superposition principle for homogeneous recurrences.
- **Nonhomogeneous Solution Structure:** State why the general solution is $a_n = a_n^{(p)} + a_n^{(h)}$.

▮ Proof Challenges & Calculations 5.3

- **Prime Factorization via Strong Induction:** Write the full inductive proof that every integer $n > 1$ can be factored into primes.
- **Superposition Proofs:**
 - Prove that if $\{a_n^{(1)}\}, \{a_n^{(2)}\}$ solve the homogeneous recurrence, then $\alpha a_n^{(1)} + \beta a_n^{(2)}$ is also a solution.
 - Prove that if $\{a_n^{(1)}\}, \{a_n^{(2)}\}$ solve the nonhomogeneous recurrence, then $\{a_n^{(1)} - a_n^{(2)}\}$ solves the homogeneous equation.
- **Characteristic Root Equivalence:** Prove that $a_n = r^n$ ($r \neq 0$) is a solution to $a_n = \sum_{i=1}^k c_i a_{n-i} \Leftrightarrow r^k - \sum_{i=1}^k c_i r^{k-i} = 0$.
- **Recurrence Calculations:**
 - Solve $a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$ with $a_0 = 2, a_1 = 5, a_2 = 15$.
 - Solve $a_n = 6a_{n-1} - 12a_{n-2} + 8a_{n-3}$ with $a_0 = 1, a_1 = 4, a_2 = 28$.

▮ Textbook Exercise Checklist.

Section	Textbook Problems	Status
Section 5.1	Problems 19, 21, 22	▫ Done

6 Counting and Combinatorics

▮ Definitions & Concepts to State 6.1

- **Basic Counting Principles:** State formulas for total functions $A \rightarrow B$ (n^m) and one-to-one functions $P(n, m)$.
- **Pigeonhole Principle:** State the basic and Generalized Pigeonhole Principle.
- **Permutations & Combinations:** State formulas for $P(n, r)$, $\binom{n}{r}$, and multinomial coefficients $\binom{n}{n_1, \dots, n_k}$.
- **Derangements:** Define derangement (D_n) and state its summation formula.

▮ Theorems & Identities to State 6.2

- **Binomial Theorem:** State $(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$.
- **Combinatorial Identities to Know:**
 - Pascal's Identity: $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$
 - Sum of Coefficients: $\sum_{k=0}^n \binom{n}{k} = 2^n$
 - Vandermonde's Identity: $\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{r-k} \binom{n}{k}$
 - Sum of Squares: $\sum_{k=0}^n \binom{n}{k}^2 = \binom{2n}{n}$
 - Hockey-Stick Identity: $\sum_{k=r}^n \binom{k}{r} = \binom{n+1}{r+1}$
- **Principle of Inclusion-Exclusion (PIE):** State the general formula for $|\bigcup_{i=1}^n A_i|$.
- **Surjective Functions Count:** State formula for onto functions from size m to size n : $\sum_{j=0}^n (-1)^j \binom{n}{j} (n-j)^m$.

¶Proof Challenges & Calculations 6.3**□ Combinatorial Proofs:**

- Provide an algebraic and a combinatorial proof for Pascal's Identity.
- Give a combinatorial proof for the Hockey-Stick Identity.
- Prove $\varphi(mn) = \varphi(m)\varphi(n)$ for $\gcd(m, n) = 1$ using PIE.

□ Stars and Bars Problems:

- Find non-negative integer triples (x_1, x_2, x_3) satisfying $x_1 + x_2 + x_3 = 10$.
- Find positive integer triples $(x_i \geq 1)$ satisfying $x_1 + x_2 + x_3 = 10$.

□ Combinatorial Problems:

- Calculate onto functions from a set of 6 elements to a set of 3 elements.
- Calculate ways 5 friends in Secret Santa can draw names such that nobody draws their own name (D_5).
- How many strictly alphabetical length-5 lists can be made from $\{A, B, C, D, E, F, G, H, I\}$?
- How many permutations of $\{A, B, C, D, E, F\}$ have D occurring before A ?
- Prove that for any $a \in \mathbb{N}$, there exist distinct $k, l \in \mathbb{N}$ such that $10 \mid (a^k - a^l)$ via Pigeonhole Principle.

7 Relations, Equivalence Relations, and Posets

▮ Definitions & Concepts to State 7.1

- **Relation Definitions:** Define relation R on set A . How many relations exist on an n -element set?
- **Relation Properties:** Write formal definitions of:
 - Reflexive
 - Symmetric
 - Antisymmetric
 - Transitive
- **Representations:** Describe zero-one matrix representation M_R and digraphs.
- **Composition & Transitive Closure:** Define $S \circ R$, Boolean product $M_R \odot M_S$, and connectivity relation R^* .
- **Equivalence Relations:** Define equivalence relation and equivalence classes $[a]$.
- **Poset Fundamentals:** Define partial ordering ($\leq$), comparability, total order (chain), and Hasse diagram.
- **Extremal Elements:** Define maximal, minimal, greatest, least, upper bound, lower bound, LUB (supremum), and GLB (infimum).

▮ Theorems & Identities to State 7.2

- **Transitivity Criterion:** State R is transitive $\Leftrightarrow R^n \subseteq R$ for all $n \geq 1 \Leftrightarrow R^2 \subseteq R$.
- **Fundamental Theorem of Equivalence Relations:** State the partition-equivalence duality theorem.
- **Equal-Class Cardinality Theorem:** State the relation $|R| = m|A|$ when all classes have size m .
- **Composition Monotonicity:** State $R_1 \subseteq R_2 \implies S \circ R_1 \subseteq S \circ R_2$.

[[Proof Challenges & Calculations 7.3

- **Properties Verification:** Determine whether $R = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : a = b + 1\}$ is antisymmetric.
- **Matrix Composition Proof:** Prove $M_{S \circ R} = M_R \odot M_S = [t_{ij}]$ where $t_{ij} = \bigvee_k (r_{ik} \wedge s_{kj})$.
- **Equal-Class Proof:** Prove that if every equivalence class of R on A has size m , then $|R| = m|A|$.
- **Poset Analysis:** In $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$:
 - Identify greatest and least elements.
 - Find upper bounds, lower bounds, LUB, and GLB for $A = \{\{1\}, \{2\}\}$.